

Misallocation, Variable Markups, and the Labour Share:

Plant-Level Evidence from Indian Manufacturing, 2012–2022*

Laxmikanta Sarangi Falguni Pattanaik

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Abstract

How much would Indian manufacturing produce if capital and labour were allocated efficiently across its plants — and who would gain? We measure within-industry misallocation on a newly built panel of roughly 39,000 organised-sector plants per year (2012–2022), first with the Hsieh–Klenow accounting and then inside a calibrated Kimball demand system in which markups vary with plant size and respond to reallocation. Four results. First, measured misallocation is large — hypothetical TFP gains from equalising marginal revenue products within industries are 121–184 per cent — but follows an inverted U, worsening to a 2016 peak and improving markedly thereafter; capital wedges dominate labour wedges in every year and every state. Second, variable markups account for under two per cent of measured revenue-productivity dispersion, yet they price away 7–19 per cent of the gains a constant-markup model promises, because reallocation’s winners face less elastic demand and re-price. Third, the same re-pricing overturns the distributional presumption: equalising marginal products *lowers* the aggregate labour share by 2.65 percentage points (against -0.12 under CES), while marginal-cost pricing raises it by 8.7 points — efficiency for workers runs through markups, not through reallocation. Fourth, the size-dependent component of wedges emphasised in recent

*Department of Humanities and Social Sciences, Indian Institute of Technology Roorkee. This paper is the second of a series based on a newly constructed plant panel from the Annual Survey of Industries; the companion papers measure markups and markdowns (working paper 1) and their macroeconomic determinants (working paper 3). ASI unit records are used under licence from the Ministry of Statistics and Programme Implementation and cannot be redistributed; all aggregate results and code are available from the authors. All errors are ours.

work on India is thin inside the organised sector (2–6 per cent of dispersion; size-neutralising gains of 1–2 per cent), consistent with size distortions binding at the formal–informal boundary below the ASI’s coverage floor. Sensitivity analysis in the spirit of Zhang–Xia moves every level and no conclusion.

JEL codes: O11, O47, D24, D61, L11, E25.

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1 Introduction

Aggregate productivity is not only a matter of what technologies an economy possesses but of how its inputs are assigned to them. When a plant’s marginal revenue product of capital exceeds its neighbour’s, moving a unit of capital between them raises output with no new technology and no new investment; an economy that leaves many such trades unexecuted is poorer than its production possibilities allow. The insight is old, but Restuccia and Rogerson (2008) and Hsieh and Klenow (2009) turned it into a measurement programme: under explicit assumptions on technology and demand, the *dispersion* of revenue productivity across plants within narrowly defined industries is exactly the footprint of the distortions — taxes, credit constraints, regulation, market power — that drive marginal products apart, and a counterfactual that equalises marginal products converts that dispersion into an aggregate TFP number. For India the programme carried a striking message: Hsieh and Klenow (2009) put the gains from full within-industry equalisation at roughly 100–130 per cent of manufacturing TFP in the 1987–1994 cross-sections — India became the canonical high-misallocation economy, and “reallocation” a standing item on its reform agenda.

This paper re-measures Indian manufacturing misallocation on a new plant panel covering 2012–2022, and asks the question the canonical accounting cannot: what happens to the promised gains — and to their distribution — when market power is allowed to respond? The question is forced by our companion paper (working paper 1), which documents on the same plants that product-market power is modest but *size-dependent*, and labour-market power pervasive. In the canonical framework every markup is a “distortion” indistinguishable from a tax, and every reallocation leaves prices unchanged. Neither is innocuous. A markup that rises with plant size is not a residual wedge; it is an equilibrium object that will move when resources move. And if reallocation’s winners — the large, productive, high-markup plants — respond to their new scale by re-pricing, then the aggregate consequences of “efficient” reallocation, for output and especially for the division of income between labour and profits, can differ sharply from the constant-markup arithmetic.

Stated as questions, the paper asks three things. **Q1 (magnitude and anatomy):** how large is within-industry misallocation in organised Indian manufacturing over 2012–2022, how has it moved, and which factor market — capital or labour — carries it? **Q2 (the markup correction):** how much of measured dispersion is markups rather than wedges, and how much of the counterfactual gain survives when markups are modelled and allowed to respond? **Q3 (incidence):** who gains from reallocation — in factor shares (does labour’s share of income rise?), across the size distribution (are the wedges a tax on size, as recent work argues for India?), and across space (which states, and which side of the rural–urban boundary, are most misallocated)? The identifying lever throughout is

the one the misallocation literature made standard: under monopolistic competition and Cobb–Douglas technology, a plant’s revenue productivity is proportional to a geometric average of its wedges — so *dispersion* in revenue productivity within an industry measures distortion, and its factor-specific components (marginal revenue products of capital and labour) attribute the distortion to factor markets. What is new here is the demand side wrapped around that lever: a calibrated Kimball system in which the markup is a known, increasing function of relative size, so that the markup component of measured dispersion can be separated from residual wedges, and the counterfactuals can let prices respond.

Answering these questions faces four well-known challenges, which discipline the design. **(i) Measurement error masquerades as misallocation.** Revenue productivity is a ratio of noisy survey variables, and [Bils et al. \(2021\)](#) show that a substantial share of measured US and Indian dispersion can be noise. We respond by trimming exactly as the literature prescribes, by privileging *comparisons* — across years, factors, states, technologies — over levels, and by reporting a gross-output reconstruction in which the same technology is held fixed and only the accounting changes. **(ii) Parameter fragility.** [Zhang and Xia \(2022\)](#) demonstrate that HK-style gains swing from 44 to 362 per cent across defensible parameterisations on Chinese data. We implement their programme rather than merely citing it: every maintained parameter — capital elasticities, the demand system, the demand parameters — is varied in Section 6, with one instructive degeneracy reported as a finding. **(iii) Markups contaminate the wedges.** In the canonical accounting a high-markup plant looks identical to a taxed plant. The Kimball structure resolves the observational equivalence *by theory*: the markup is pinned down by the plant’s relative size through an estimated demand curve, and only the remainder of revenue productivity is a wedge. **(iv) The counterfactual itself is model-dependent.** Because Kimball demand is non-CES, the liberalised equilibrium cannot be computed plant by plant; it is a fixed point in each industry’s demand index. We follow the numerical implementation of [Liang \(2023\)](#) — whose replication code we execute verbatim on the Indian panel — and validate our pipeline against her published results decade-for-decade.

We bring this machinery to the panel described in the companion paper: from the raw ASI block files we construct 481,322 plant-years (2012–2022) with value added, capital, and wage bills, of which 429,534 survive the framework’s cleaning — about 39,000 plants per year, every organised manufacturing industry, every state. Four sets of findings emerge.

First, large but improving misallocation, carried by capital. The variance of within-industry log revenue productivity falls from 0.73 (2012) to 0.59 (2022) — high by international standards throughout, but on a clear downward path that accelerates after 2018. The factor anatomy is lopsided: the dispersion of the marginal revenue product of capital (s.d. ≈ 1.1 – 1.2) is half again that of labour (s.d. ≈ 0.7 – 0.8) in every year, every state, and both sectors of the rural–urban divide. Counterfactual gains from full within-industry

equalisation are 121–184 per cent across annually recalibrated models — above [Liang \(2023\)](#)’s 92–110 per cent for the 2000s, joining her series smoothly at the overlap — and trace an inverted U that peaks in 2016 and falls steeply after 2020. *Second, markups are a small share of the disease and a large tax on the cure.* Modelled markups account for only ≈ 1.6 per cent of revenue-productivity dispersion — residual capital and output wedges dominate — yet the Kimball market counterfactual delivers only 81–93 per cent of the CES-promised gain: reallocation concentrates sales in plants whose demand is less elastic, and their re-pricing “taxes away” 7–19 per cent of the constant-markup arithmetic. *Third, the labour share falls when allocation improves.* Equalising marginal products lowers the aggregate labour share by 2.65 percentage points in the Kimball economy; freezing markups at a constant (CES) makes the same experiment distributionally neutral (−0.12 points). The comparison isolates the mechanism — it is the *endogeneity* of markups, not their heterogeneity, that turns reallocation against labour — and the planner benchmark shows the converse: marginal-cost pricing raises labour’s share by 8.7 points before any reallocation. *Fourth, size-dependent wedges are a formality-boundary phenomenon.* Transplanting the diagnostics of [Hsieh et al. \(2026\)](#) into the organised sector, plant size explains only 2–6 per cent of revenue-productivity dispersion; neutralising the size-fitted component of wedges yields output gains of 1.2–1.9 per cent — under one per cent of the full-equalisation benchmark computed with the same machinery — and the largest plants are, if anything, wedge-favoured. India’s tax on size, and the efficiency–inequality trade-off it generates, lives at the formal–informal margin below the ASI’s floor, not among the factories studied here.

The paper contributes to three literatures. To the *misallocation measurement* literature it adds the first annually recalibrated variable-markup accounting for India — eleven cross-sections, each with its own demand calibration — turning the single-year snapshots of the prior literature into a time path, and it implements the [Zhang and Xia \(2022\)](#) sensitivity programme end-to-end on Indian data, including a degeneracy (domestic cost-share elasticities make the demand system uncalibratable) that vindicates [Hsieh and Klenow](#)’s original argument for external elasticities in an unexpected way. To the *markups and macro* literature following [Edmond et al. \(2015\)](#), [Peters \(2020\)](#) and [Edmond et al. \(2023\)](#), it contributes a sharp empirical decomposition of the two roles of markups — negligible in the variance accounting, decisive in the counterfactual — and the labour-share reversal, which extends [Zhang and Xia](#)’s China finding from a dampening to a sign flip once markups re-optimize. To the *Indian development* literature — [Hsieh and Klenow \(2009\)](#), [Bollard et al. \(2013\)](#), [García-Santana and Pijoan-Mas \(2014\)](#), [Duranton et al. \(2015\)](#), [Hsieh et al. \(2026\)](#) — it updates the central estimate a decade forward on a fresh panel, relocates the size-dependent component of the problem to the formality boundary, and documents that the much-invoked reallocation reform would, within the organised sector as it stands, be a reform against labour’s share of income.

Table 1 collects the headline numbers. Section 2 situates the paper; Section 3 describes the data; Section 4 develops the framework equation by equation; Section 5 presents results; Section 6 the sensitivity analysis; Section 7 draws the interpretation together; Section 8 concludes. Appendices document the sample (A), the numerical implementation (B), the full state panel (C), and the gross-output reconstruction (D).

Table 1: Findings at a glance (2012–2022 unless noted; “gain” = counterfactual TFP gain from within-industry marginal-product equalisation).

Object	Estimate
Estimation sample	429,534 plant-years ($\approx 39,000/\text{year}$)
$\text{var}(\log \text{TFPR}) \times 100$, within NIC-4	72.9 (2012) $\rightarrow$ 58.6 (2022)
s.d. MRPK vs s.d. MRPL (2017)	1.15 vs 0.74 — capital wedges dominate
Markup share of TFPR variance	$\approx 1.6\%$ (baseline); 0.4% (own- α)
Kimball market gain (2017 reference)	175%; planner 179%; CES 189%
Per-year gains (recalibrated)	121–184%, inverted U peaking 2016
Kimball/CES gain ratio	0.81–0.93 — markups tax the cure 7–19%
Δ labour share, equalisation	–2.65pp (Kimball) vs –0.12pp (CES)
Δ labour share, marginal-cost pricing	+8.7pp (before reallocation)
Size-fitted share of TFPR variance	2.2% (2012) $\rightarrow$ 5.6% (2022)
Size-neutralising counterfactual gain	1.2–1.9% per year
Most/least misallocated states	Odisha, AP, WB (>75) / Punjab, Haryana (≈ 49)

2 Related literature

The accounting strand. Restuccia and Rogerson (2008) established in a calibrated heterogeneous-plant model that idiosyncratic “taxes” correlated with productivity can depress aggregate TFP by 30–50 per cent; Hsieh and Klenow (2009) inverted the logic into measurement, showing that under CES demand and Cobb–Douglas technology the within-industry dispersion of revenue productivity (TFPR) is a sufficient statistic for the TFP cost of distortions, and estimating gains of 86–115 per cent for China and roughly 100–130 per cent for India. The framework’s power is its parsimony — two wedges per plant, recovered from two ratios — and its weaknesses are the flip side: everything that moves revenue productivity that is *not* a policy distortion (markups, adjustment costs, risk, measurement error) is scored as misallocation. Restuccia and Rogerson (2017) survey the ensuing debate; Bergquist et al. (2026) give the modern general statement of when marginal-product equalisation is, and is not, the right welfare benchmark.

The critique strand. Three critiques matter for our design. Asker et al. (2014) show capital adjustment costs generate TFPR dispersion in a frictionless-on-average world — part of any capital-heavy dispersion is transition, not tax. Bils et al. (2021) bound the contribution of additive measurement error using the covariance between revenue growth and input growth; their correction shrinks but does not eliminate the US–India gap. And

Zhang and Xia (2022) attack parameter fragility directly: on Chinese data, defensible combinations of demand elasticity and returns to scale move the headline gain by a factor of eight, leading them to advocate domestic estimation and systematic sensitivity reporting — the protocol Section 6 follows. We treat all three not as reasons to abandon the accounting but as reasons to privilege its *comparative* statements: across factors, years, places, and model variants, all of which difference out level-shifting pathologies.

The variable-markup strand. Once markups vary across plants, they enter TFPR and the CES accounting mislabels them. Edmond et al. (2015) and Edmond et al. (2023) quantify markup misallocation in trade and macro settings; Peters (2020) makes markup dispersion an equilibrium object of entry and growth. Closest to us, Liang (2023) embeds the HK accounting in a Kimball–Klenow and Willis (2016) demand system calibrated on the Indian ASI for 1998–2008 and shows the gains promised by CES overstate what a variable-markup economy can deliver. Methodologically this paper is her decade-forward extension — her code, our new panel, eleven annual recalibrations — plus three additions: the labour-share counterfactual with endogenous re-pricing, the triangulation of Kimball markups against production-side and oligopoly (Atkeson and Burstein, 2008) alternatives on the same plants, and the Hsieh et al. (2026) size diagnostics.

The India strand. Bollard et al. (2013) document the puzzle of strong within-plant productivity growth alongside stagnant allocation; García-Santana and Pijoan-Mas (2014) attribute part of measured misallocation to the small-scale reservation laws (largely dismantled by our sample period); Duranton et al. (2015) emphasise land and infrastructure. Most recently, Hsieh et al. (2026), on combined formal-plus-informal data for 2000–2012, argue that the *size-dependent* component of Indian wedges is what reform can realistically target, and that removing it would raise output little (1–4 per cent) while raising the skill premium and lowering welfare — misallocation as the price of equality. Our size diagnostics engage their argument on the organised sector directly, and our labour-share counterfactual delivers a complementary caution through an entirely different channel: theirs runs through skill composition, ours through markups; both warn that reallocation’s efficiency gains and workers’ gains are different objects.

3 Data

3.1 Source and sample

The data are the plant panel constructed in working paper 1 from the raw block-level files of the Annual Survey of Industries with common factory identifiers: 481,322 plant-years over 2012–2022 with the value-added objects this framework requires. The framework uses *nominal* values — every comparison it makes is within an industry-year cell, so industry-year price levels cancel by construction; this is the standard HK justification and it spares

the analysis any deflator error. Plant value added is ex-factory output less materials; capital is the closing net book value of fixed assets; labour input is the wage bill (total emoluments), so that cross-plant differences in hours, skill and effort are absorbed into efficiency units — an important choice, since with headcount labour a plant employing better-paid, more productive workers would spuriously show a high marginal product of labour.

3.2 Cleaning and the estimation sample

The cleaning follows [Liang \(2023\)](#) line by line so that results are comparable decade to decade: drop plant-years with non-positive value added, output, capital, employment, wage bill, or materials; retain industry-year (NIC-4 $\times$ year) cells with at least fifty plants; recover the model objects; trim the one per cent tails of the log deviations of TFPR and TFPQ from industry aggregates, by year. The estimation sample is **429,534** plant-years — roughly 39,000 per year, comparable in density to Liang’s 30–40,000 — spanning every organised manufacturing division and every state. Three unavoidable deviations from her implementation are documented in [Appendix A](#) with their expected direction of effect (closing rather than average capital; energy and services not netted from value added; no profit-based filter); none affects within-industry-year dispersion in any signed way we can detect.

3.3 Parameters taken from outside the data

The industry capital elasticities α_s are taken from the U.S. NBER–CES manufacturing database, exactly as in [Hsieh and Klenow \(2009\)](#) and [Liang \(2023\)](#), on the argument — which [Section 6](#) demonstrates empirically rather than merely asserts — that domestic factor shares confound technology with the very distortions under study. Liang’s elasticities are coded in the NIC-2004 vintage and mapped to NIC-2008 through the official division-level concordance. The rental rate is $r = 0.10$ (five per cent real rate plus five per cent depreciation, as in HK); the wage is normalised to one (only relative marginal products matter within cells). One flag for later: the tobacco division inherits $\alpha_s = 0.95$ from the near-zero U.S. tobacco labour share — the most extreme value in the mapping — and is retained in the baseline for comparability but examined separately in [Section 6](#).

4 Framework

This section develops the full chain from data to counterfactual, equation by equation: the technology and wedges [\(4.1\)](#), the demand system that lets markups vary [\(4.2\)](#), the identification of its two parameters [\(4.3\)](#), the counterfactual equilibria [\(4.4\)](#), and the size-wedge diagnostics [\(4.5\)](#).

4.1 Technology, wedges, and revenue productivity

The economy in year t (the model is solved year by year; time subscripts are suppressed) consists of S four-digit industries; industry s contains n_s monopolistically competitive plants. Factors are supplied inelastically and are sector-specific, so all action is *within*-industry reallocation — the conservative choice, since cross-industry wedges would only add gains. Plants produce with the value-added technology

$$y_{si} = A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}, \quad (1)$$

where A_{si} is physical productivity (TFPQ), K is capital, and L is labour in efficiency units (the wage bill). Equation (1) is deliberately spare: Cobb–Douglas with a common α_s within the industry means that *any* cross-plant difference in factor proportions must be explained by prices, wedges, or market power — not technology. That is what gives the accounting its bite, and what the robustness section must therefore stress-test.

Each plant faces two idiosyncratic wedges, following HK: a capital distortion $\tau_{K,si}$ scaling the relative cost of capital, and an output distortion $\tau_{Y,si}$ scaling revenue. Here they are *residual* distortions — what remains after markups are modelled explicitly in the next subsection. The plant maximises profit subject to its demand curve, and the first-order conditions invert into recovery formulas that use only observables and parameters:

$$\tau_{K,si} = \frac{\alpha_s}{1 - \alpha_s} \cdot \frac{wL_{si}}{rK_{si}}, \quad \tau_{Y,si} = \mu_{si} \cdot \frac{wL_{si}}{(1 - \alpha_s) P_{si} y_{si}}, \quad (2)$$

with μ_{si} the plant's markup. Read (2) as diagnostics. The capital wedge is identified from factor *proportions*: under (1) an undistorted plant spends $\alpha_s/(1 - \alpha_s)$ times its wage bill on capital services, so a plant whose wage-bill-to-capital-cost ratio exceeds that benchmark behaves as if capital were expensive to it — $\tau_K > 1$ — whatever the reason (credit rationing, land, uncertainty). The output wedge is identified from the *level* of the labour share relative to the markup-adjusted benchmark: a plant whose revenue is high relative to its factor payments, beyond what its markup explains, behaves as if subsidised. Undistorted plants have $\tau_K = \tau_Y = 1$.

Revenue productivity — TFPR, the central measured object — is price times physical productivity, and combining (1)–(2) with the demand system gives

$$\text{TFPR}_{si} = P_{si} A_{si} = \mu_{si} (r \tau_{K,si})^{\alpha_s} w^{1-\alpha_s} \tau_{Y,si} \propto \frac{P_{si} y_{si}}{K_{si}^{\alpha_s} (wL_{si})^{1-\alpha_s}}, \quad (3)$$

$$\text{TFPQ}_{si} \propto \frac{(P_{si} y_{si})^{\frac{\sigma}{\sigma-1}}}{K_{si}^{\alpha_s} (wL_{si})^{1-\alpha_s}}. \quad (4)$$

Equation (3) is the lever of the whole literature and deserves its intuition stated plainly.

In an undistorted industry, TFPR is *equalised* across plants no matter how unequal their physical productivities: a more productive plant expands, and under downward-sloping demand its price falls, until price times productivity is the same as everyone else’s — high A shows up as high output, not high TFPR. Therefore any *dispersion* in TFPR within an industry is direct evidence that something — μ , τ_K , or τ_Y in (3) — prevents marginal products from equalising. The factor-specific versions

$$\text{MRPL}_{si} \propto \frac{P_{si}y_{si}}{wL_{si}}, \quad \text{MRPK}_{si} \propto \frac{P_{si}y_{si}}{K_{si}}, \quad (5)$$

attribute the dispersion: a plant with high MRPK relative to its industry is starved of capital; one with high MRPL is starved of labour. Equation (4) inverts revenue into quantity using the demand elasticity ($y = (Py)^{\sigma/(\sigma-1)}$ up to industry constants), which is what allows physical productivity to be recovered from revenue data at all — and is also why everything downstream depends on the demand system, the subject of the next subsection.

Aggregation and the counterfactual follow HK’s equations 15–16. Industry TFP under the actual allocation, and under the efficient one, are

$$A_s^{\text{actual}} = \left[\sum_{i \in s} \left(A_{si} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}, \quad A_s^{\text{efficient}} = \left[\sum_{i \in s} A_{si}^{\sigma-1} \right]^{\frac{1}{\sigma-1}}, \quad (6)$$

with $\overline{\text{TFPR}}_s$ the appropriately (harmonically, VA-share) weighted industry average; manufacturing TFP aggregates across industries Cobb–Douglas with value-added weights η_s , and the **gain** is $\prod_s (A_s^{\text{eff}}/A_s^{\text{act}})^{\eta_s} - 1$. The structure of (6) explains two properties used repeatedly below. A plant contributes to the loss according to the *ratio* of its TFPR to the industry mean — so it is dispersion, not levels, that costs output. And under joint log-normality the loss takes the transparent form $\ln(A^{\text{eff}}/A^{\text{act}}) \approx \frac{\sigma}{2} \text{var}(\ln \text{TFPR})$: the cost is *convex in the demand elasticity* — more substitutable varieties mean bigger losses from the same dispersion, because the efficient allocation would concentrate production more aggressively — which is why the calibration of the demand system in Section 4.3 matters as much as the measurement of dispersion itself.

4.2 From CES to Kimball: letting the markup vary

Under CES demand every plant charges $\sigma/(\sigma-1)$: markups are a constant, they difference out of TFPR dispersion, and reallocation leaves pricing unchanged. Both implications are counterfactual — our companion paper measures a positive size–markup gradient on these very plants — so we replace CES with homothetic Kimball (1995) preferences within each

sector: the sectoral aggregate Y_s is defined implicitly by

$$\frac{1}{n_s} \sum_{i \in s} \Upsilon\left(\frac{n_s y_{si}}{Y_s}\right) = 1, \quad (7)$$

where Υ is increasing and concave with $\Upsilon(1) = 1$. Concavity of Υ is the economics: the marginal contribution of a variety to the sector aggregate diminishes in its relative size, so demand curves get *steeper* for plants that are large relative to their market. We adopt the [Klenow and Willis \(2016\)](#) specification, under which the demand elasticity facing a plant with relative quantity $x_{si} = n_s y_{si} / Y_s$ is

$$\sigma(x_{si}) = \varsigma x_{si}^{-\varepsilon/\varsigma}, \quad (8)$$

governed by two parameters with clean separate jobs: ς sets the *level* of demand elasticities (the CES/HK case is the limit $\varepsilon \rightarrow 0$), and the **superelasticity** ε/ς sets the *rate* at which a plant's elasticity falls — hence its markup rises — as its relative size grows. The plant's markup follows from the Lerner condition,

$$\mu_{si} = \frac{\sigma(x_{si})}{\sigma(x_{si}) - 1}, \quad (9)$$

so markups are no longer free parameters or residuals: they are a known increasing function of relative size. This single change does three things to the accounting. It *decontaminates the wedges*: the μ_{si} term in (2)–(3) is now measured, so τ_Y is a residual distortion net of market power rather than a mixture. It *disciplines the counterfactual*: when reallocation changes a plant's relative size, (8)–(9) say exactly how its markup responds — the incomplete pass-through channel that turns out to drive the distributional results. And it *creates an estimating equation*: (9) implies a log-linear relation between a transform of the plant's profit margin and its market share, which identifies the superelasticity from panel data (next subsection).

4.3 Identifying the two demand parameters

The demand system has two parameters per year, identified from two moments, both following [Liang \(2023\)](#) exactly.

Moment 1: the aggregate profit margin pins the elasticity level. The aggregate Lerner markup

$$\mu^{agg} = \frac{\sum_{si} P_{si} y_{si}}{\sum_{si} w L_{si} / (1 - \alpha_s)} \implies \bar{\sigma}^{\text{target}} = \frac{1}{1 - 1/\mu^{agg}} \quad (10)$$

is a model-free accounting object: aggregate value added over aggregate implied factor payments. Equating it to the margin of a representative firm facing the sales-weighted

average demand elasticity gives the target $\bar{\sigma}$; the parameter ς is then solved — by an outer secant loop around the full within-industry equilibrium computation — so that the model’s sales-weighted average elasticity hits the target. The logic is worth pausing on: a decade in which manufacturing margins compress is, through (10), a decade in which calibrated demand becomes more elastic, and through the convexity of the loss in σ , a decade in which the *same* measured dispersion costs more output. The calibration table in Section 5.3 shows precisely this co-movement, and it is why we recalibrate annually instead of importing one year’s parameters into another’s dispersion.

Moment 2: the within-plant margin–share slope pins the superelasticity.

The Klenow–Willis structure implies the estimating equation

$$\frac{1}{\mu_{sit}} + \ln\left(1 - \frac{1}{\mu_{sit}}\right) = \text{const} - \frac{\varepsilon}{\varsigma} \ln \lambda_{sit} + \text{plant FE} + \text{year FE} + e_{sit}, \quad (11)$$

where μ_{sit} is the plant’s accounting Lerner markup and λ_{sit} its within-industry market share. The left side is the exact transform under which Klenow–Willis demand is log-linear in the share; plant fixed effects confine identification to *within-plant* variation — a plant whose share grows must, under (8), show a predictably fatter margin — which guards against cross-sectional confounds (better plants having both high shares and high margins for technological reasons). On 237,064 usable plant-years,

$$\widehat{\varepsilon/\varsigma} = 0.183 \quad (\text{s.e. } 0.005), \quad (12)$$

against Liang’s 0.14 (s.e. 0.005) for 2001–2008: the same order, indicating somewhat steeper size-dependence of markups in the later decade. Three independent checks corroborate it: Liang reports a production-side alternative yields 0.15 in her data; our companion paper’s production-side size gradient is positive within plants; and the calibrated markup distribution (below) matches the production-side translog distribution in level and shape. Following Liang, the superelasticity is held at its pooled value across reference years, so annual recalibration operates through ς alone.

The calibrated model at the 2017 reference year. Solving on the mid-sample cross-section, the outer loop converges in five iterations to $\hat{\varsigma} = 5.34$ (so $\varepsilon = 0.98$), with the model’s sales-weighted elasticity within 10^{-3} of the 4.457 target and the implied aggregate markup 1.315 against the data’s 1.289. The equilibrium objects are economically sensible: firm demand elasticities have a 10th percentile of 4.29, median 6.87, 90th of 12.13 (Liang’s Table 1: 3.17–7.12); Kimball markups have mean 1.186, median 1.170, and a 99th percentile of 1.461 — right-skewed with a long thin upper tail, and comparable to the production-side translog markup median of 1.241 for the same plants and year. That two entirely different identification strategies — production-side cost shares there, demand-side market-share moments here — deliver overlapping markup distribu-

tions for the same plants is among the strongest internal-validity results of the project, and Section 5.4 pushes the comparison to plant level.

4.4 Four equilibria and what their differences isolate

The **liberalisation** counterfactual sets $\tau_{K,si} = \tau_{Y,si} = 1$ for all plants — equalising marginal revenue products of both factors within each industry — and re-solves the industry equilibrium. Because demand is Kimball, this is not a formula but a fixed point: each plant’s markup, price, and relative output depend on the sectoral demand index, and the index must satisfy (7). The numerical implementation is Liang’s own — secant iterations on the demand index within the calibration loop over ς , her Mata root-finders executed verbatim — with one engineering addition documented in Appendix B. Aggregate effects are Cobb–Douglas across industries with value-added weights. Four equilibria are computed:

- **Baseline:** actual wedges, Kimball pricing — the economy as measured.
- **Kimball market:** wedges removed, plants *re-optimize* markups as their relative sizes change. This is the honest liberalisation: reallocation happens, and its winners re-price along (8)–(9).
- **Kimball planner:** wedges removed *and* marginal-cost pricing imposed. The gap to the market equilibrium isolates the incomplete-pass-through channel — what variable markups cost on top of the wedges.
- **CES market:** wedges removed with markups frozen at $\sigma/(\sigma - 1)$, calibrated to the same average elasticity — the HK benchmark, what a constant-markup model *promises* from the same data.

The pairwise differences are the paper’s decomposition: Kimball-market versus baseline is the attainable gain; CES versus Kimball-market is the “markup tax” on the cure; planner versus market is pass-through; and the labour-share versions of the same four equilibria (Section 5.5) turn the decomposition distributional.

4.5 Size-dependent wedges: the HKKR diagnostics

Hsieh et al. (2026) argue that the policy-relevant component of Indian wedges is the part that varies systematically with plant *size* — size-dependent regulation, enforcement, and informality — and that removing it trades efficiency against equality. Three of their diagnostics transplant onto our sample (at their $\sigma = 3$). First, the size gradient: a revenue-weighted regression of the within-industry log-TFPR deviation on their seven

employment bins with NIC-4×year effects. Second, the variance decomposition

$$\text{size share}_t = \frac{\text{var}(\widehat{\text{fit}}_{it})}{\text{var}(\ln \widetilde{\text{TFPR}}_{it})}, \quad (13)$$

the share of within-industry TFPR variance explained by the size-bin fit. Third, the **size-neutralising counterfactual**: remove only the size-fitted component of wedges (mean-preserving within industry) and recompute (6) — their Section 5.1 exercise, answering “what would fixing size-dependent policy alone be worth?” against the full-equalisation benchmark computed with the same machinery.

5 Results

5.1 The anatomy of dispersion

Table 2 reports the within-industry dispersion of log TFPR and its two factor components, year by year, on the US- α value-added pipeline.

Table 2: Within-industry (NIC-4×year) dispersion of revenue productivity and factor marginal products, by year. Standard deviations of log deviations from industry aggregates; final column is $\text{var}(\log \text{TFPR}) \times 100$.

Year	N	s.d. MRPL	s.d. MRPK	s.d. TFPR	$\text{var} \times 100$
2012	34,595	0.806	1.195	0.854	72.9
2013	36,152	0.783	1.199	0.849	72.0
2014	37,942	0.763	1.191	0.830	68.9
2015	37,550	0.760	1.177	0.826	68.3
2016	39,776	0.742	1.159	0.805	64.7
2017	39,305	0.736	1.146	0.798	63.7
2018	39,426	0.709	1.113	0.766	58.6
2019	39,817	0.721	1.120	0.775	60.0
2020	40,830	0.722	1.112	0.776	60.2
2021	41,519	0.713	1.101	0.764	58.4
2022	42,622	0.710	1.107	0.766	58.6

Three facts organise the table. **First, the level.** A TFPR variance of 0.59–0.73 means the plant at one standard deviation above its industry mean has revenue productivity more than twice the plant one standard deviation below — enormous unexecuted trades by any international standard, and the raw material for the large counterfactual gains below. **Second, the trend.** Dispersion falls monotonically-with-wiggles from 72.9 (2012) to 58.6 (2022) — a fifth of measured misallocation gone in a decade. The decline is not an artefact of sample growth (the cell structure is fixed and N rises smoothly) and survives every re-measurement in Section 6. **Third, the factor anatomy.** The dispersion of MRPK is half again that of MRPL in every single year — around 1.15 against 0.74 — so

the Indian allocation problem is first a *capital*-market problem: credit, collateral, land, and vintage frictions that drive plants’ shadow costs of capital apart. The labour margin is tighter but far from clean, and our companion paper’s finding of pervasive monopsony gives part of that residual a name and a mechanism. Within this dispersion, the modelled Kimball markups account for only ≈ 1.6 per cent of TFPR variance (0.4 per cent under domestic elasticities): consistent with the companion paper’s finding that markups are modest and their dispersion tight, market power is a *small share of the disease* — though, as Section 5.3 shows, a large tax on the cure.

5.2 Calibration by year

Table 3 reports the two calibration inputs for each annual model: the aggregate Lerner markup (10) and the implied elasticity target.

Table 3: Annual calibration targets: aggregate Lerner markup and implied sales-weighted demand-elasticity target. Comparison: Liang (2023), 2004: markup ≈ 1.52 , target 2.92.

Year	N (cleaned)	Aggregate Lerner markup	Target elasticity $\bar{\sigma}$
2012	34,595	1.565	2.77
2013	36,152	1.430	3.33
2014	37,942	1.370	3.70
2015	37,550	1.389	3.57
2016	39,776	1.289	4.46
2017	39,305	1.289	4.46
2018	39,426	1.279	4.58
2019	39,817	1.190	6.27
2020	40,830	1.272	4.68
2021	41,519	1.378	3.64
2022	42,622	1.356	3.81

The margin compression of the mid-decade is clearly visible — the aggregate Lerner markup falls from 1.56 (2012) to 1.28–1.29 (2016–2018) before partially recovering — and maps through (10) into elasticity targets rising from 2.8 toward 4.6. The year 2019 is flagged now and throughout: its Lerner markup of 1.19 implies a target of 6.27, the edge of the model class’s solvable domain (Section 6); its calibration ($\hat{\varsigma} = 7.4$) and gains are reported but read as a boundary case, not evidence.

5.3 Counterfactual gains: level, path, and the markup tax

Table 4 is the paper’s central table: for each reference year, the fully recalibrated model’s TFP gains from within-industry marginal-product equalisation, under the three counterfactual equilibria of Section 4.4.

Four observations organise the table. **First, the level.** Gains of 121–184 per cent exceed Liang (2023)’s 92–110 per cent for the 2000s, in one-to-one correspondence with

Table 4: Counterfactual TFP gains by reference year (per cent), each year fully recalibrated. “Market” lets markups re-optimize; “Planner” imposes marginal-cost pricing; “CES” freezes markups (HK benchmark). Final column: attainable share of the CES promise. 2019 is a boundary case (see text).

	Ref. year	$\hat{\varsigma}$	N	Kimball market	Kimball planner	CES	Market/CES
	2012	3.45	32,324	120.8	128.7	148.4	0.81
	2013	4.09	33,540	140.8	146.2	163.8	0.86
	2014	4.53	35,595	158.5	163.2	186.8	0.85
	2015	4.34	35,226	153.3	158.2	166.9	0.92
	2016	5.36	36,601	184.2	188.1	201.4	0.91
	2017	5.34	37,699	175.1	178.7	188.6	0.93
2017 (excl. tobacco)	4.94	37,303		158.3	162.1	169.8	0.93
	2018	5.47	38,293	161.6	165.2	181.5	0.89
	2019 [†]	7.40	38,476	270.0	274.1	311.2	0.87
	2020	5.58	38,465	170.1	173.7	196.0	0.87
	2021	4.37	40,312	126.3	130.3	142.8	0.88
	2022	4.58	40,746	128.9	132.2	140.0	0.92

the larger measured dispersion; the 2012 estimate ($\hat{\varsigma} = 3.45$, gain 121%) joins her 2008 endpoint ($\hat{\varsigma} = 3.64$, gain 92%) smoothly, which is the cleanest external validation of the pipeline available. As always in this literature the level is an upper bound twice over — it prices complete elimination of within-industry wedges, and part of measured dispersion is noise and adjustment (Section 6) — so the robust content lies in comparisons. **Second, the path.** An inverted U: rising steeply to a 2016 peak (184%), declining gently to 2018, sharply after 2020, so 2021–22 gains sit near the 2012 level. Measured allocative efficiency worsened through the mid-decade — the demonetisation and GST-introduction window — and improved markedly through the COVID recovery, when exit and consolidation were substantial. The accounting cannot adjudicate causes, but the fact itself is new: it is invisible to the single-cross-section design of the prior literature, and it cautions against reading any one year’s “gain” as a structural constant. **Third, the markup tax on the cure.** The Kimball market counterfactual delivers 81–93 per cent of the CES gain: variable markups price away 7–19 per cent of what a constant-markup model promises, against a ≈ 20 per cent bite in Liang’s decade. The decomposition within the Kimball model attributes most of the remaining market–planner gap (3–8 points) to the diminishing-returns property of Kimball demand rather than to pass-through per se. **Fourth, gains and calibration co-move.** Years of high measured misallocation are years of low margins and high $\hat{\varsigma}$ (compare Tables 3 and 4): the model’s account is internally coherent — intense competition coexisting with badly allocated factors — but the co-movement also flags that part of the year-to-year variation in gains is calibration variation, one more reason to privilege the path’s shape over its points.

5.4 Whose markups? A three-way triangulation

The framework’s markups are demand-side objects. Before trusting them in counterfactuals, Section 4.3’s distribution-level agreement with the production side is pushed to plant level: for the 2017 cross-section we compute, for every plant, the markup under three identification strategies — the calibrated Kimball model; the Atkeson and Burstein (2008) oligopoly-CES model (markup from within-industry market share under nested CES); and the production-side De Loecker–Warzynski estimate from the companion paper, matched for 33,497 of the 37,699 plants.

Table 5: Markups under three identification strategies, 2017 cross-section ($N = 37,699$; 33,497 matched to a production-side estimate). Source: `markup_three_methods_2017.csv`.

	Kimball (demand)	Atkeson–Burstein (oligopoly)	DLW (production)
Mean	1.186	≈ 1.263	≈ 1.30
Std. deviation	0.087	0.013	0.244
Spearman w/ Kimball	1	0.83	0.08
Spearman w/ AB	0.83	1	0.11

Two results, one reassuring and one instructive. The reassuring one: the two *demand-side* estimators, built on different microfoundations (preferences versus conduct), agree strongly on the ranking of plants (Spearman 0.83) — they identify the same plants as the high-markup plants, exactly what one hopes of two theory-consistent measures of one latent object. Both correlate only weakly with the production-side DLW markup (0.08 and 0.11), whose dispersion (0.244) is roughly three times the Kimball and nearly twenty times the Atkeson–Burstein. This is not a defect but confirmation of the methodological point in Liang (2023): the production approach conflates the markup with the plant’s idiosyncratic distortions, so it is far more dispersed and its ranking reflects distortions as much as markups — the very features that make it right for the descriptive market-power portrait of the companion paper and wrong for counterfactual reallocation exercises. The instructive result is the *compression* of the oligopoly markups around the CES value $4.81/3.81 = 1.262$: in Atkeson–Burstein, pricing power requires a large market share, and in the fragmented four-digit industries of the ASI — where the median plant’s share is minuscule — almost every plant is atomistic. Kimball generates size-dependent markups from preferences without requiring concentration Indian manufacturing does not exhibit, which is exactly why it is the workhorse here. All three estimators agree on the aggregate level (1.19–1.30) and on markups rising with size (the AB markup–share elasticity is +0.003, tiny but precise, $t = 51$).

5.5 The labour-share counterfactual: who gains from reallocation?

The framework prices reallocation in output. Its distributional question — posed sharply by Zhang and Xia (2022) and by Hsieh et al. (2026) from different directions — is answered by recomputing the aggregate labour share, $\sum wL / \sum Py$, in each of the four equilibria on the 2017 reference year.

Table 6: The aggregate labour share under counterfactuals, 2017 reference year. Source: `labor_share_counterfactual_ref2017.csv`.

Equilibrium	Labour share	Δ vs baseline
Baseline (actual wedges, Kimball pricing)	0.343	—
Marginal products equalised, Kimball re-pricing	0.316	−2.65 pp
Marginal products equalised, CES (markups frozen) ^a	0.349	−0.12 pp
Marginal-cost pricing, actual wedges (planner)	0.430	+8.7 pp
Marginal-cost pricing, equalised (planner)	0.450	+10.8 pp

^a CES delta relative to the CES-recalibrated baseline share (0.351).

The result reverses the natural presumption, and the chain has three links worth stating because each is verifiable in the model’s algebra. *Link one: reallocation selects the strong.* Equalising marginal products moves factors toward high-TFPQ plants — under the size–markup gradient, disproportionately plants that already command large shares and charge high markups. *Link two: the winners re-price.* Under (8), a plant that grows faces less elastic demand; the winners of reallocation *raise* their markups. A higher markup is, mechanically, a lower wage–bill share of the plant’s own revenue: the plant retains as profit part of what marginal-cost pricing would deliver to factors. *Link three: composition compounds it.* The counterfactual simultaneously shifts the economy’s weight toward exactly these high-markup plants, so the aggregate share falls by more than any plant’s own change. The CES row isolates the mechanism: with markups frozen, the identical reallocation moves the labour share by −0.12 points — statistically nothing. *The entire distributional effect operates through the endogenous markup margin.* Conversely, the planner rows show what does raise labour’s share: marginal-cost pricing lifts it 8.7 points before any reallocation, 10.8 with it. Removing markups is the pro-labour experiment; removing the other wedges, in an economy where markups respond, is not. Zhang and Xia (2022) found for China that markup *heterogeneity* (held fixed) cuts the labour-share gain from liberalisation by a third (+7.4 against +11.4 points); our result sharpens theirs in a diagnosable way — when the markup distribution is allowed to *respond* rather than merely differ, the sign flips. It is the endogeneity, not the heterogeneity, that turns reallocation against labour.

5.6 The size dimension: HKKR diagnostics on the organised sector

Table 7 reports the variance decomposition (13) and the size-neutralising counterfactual, year by year; Table 8 the pooled size gradient itself.

Table 7: Size-dependent wedges, year by year: share of within-industry TFPR variance explained by seven employment-size bins, and the size-neutralising counterfactual gain against the full-equalisation benchmark (both at $\sigma = 3$, HKKR machinery).

Year	Variance decomposition ($\times 100$)			Counterfactual (%)	
	var(TFPR)	size-fitted	share (%)	size-only gain	% of full gain
2012	57.9	1.27	2.2	1.67	0.84
2013	59.1	2.08	3.5	1.78	0.89
2014	54.2	1.90	3.5	1.17	0.61
2015	57.2	2.57	4.5	1.69	0.87
2016	56.7	2.43	4.3	1.37	0.72
2017	54.9	3.04	5.5	1.55	0.83
2018	50.1	2.54	5.1	1.31	0.73
2019	49.4	2.75	5.6	1.90	1.04
2020	49.4	2.57	5.2	1.57	0.87
2021	47.3	2.34	4.9	1.29	0.72
2022	46.3	2.59	5.6	1.34	0.76

Table 8: The TFPR size gradient, pooled 2012–2022: revenue-weighted regression of within-industry log-TFPR deviation on employment-size bins (base: smallest bin), NIC-4 $\times$ year effects. Bin 7 = 500+ workers.

	Bin 2	Bin 3	Bin 4	Bin 5	Bin 6	Bin 7
Coefficient	+0.016	+0.055	+0.041	-0.015	-0.090	-0.337
(s.e.)	(0.034)	(0.032)	(0.031)	(0.031)	(0.031)	(0.031)

The reading is consistent across all three diagnostics. The size-systematic share of organised-sector misallocation is *small* — 2.2 per cent of within-industry TFPR variance in 2012, rising to 5.6 by 2022 — and the size-neutralising counterfactual yields gains of 1.2–1.9 per cent per year, strikingly close to HKKR’s own 1–4 per cent despite the different sample, and under one per cent of the full-equalisation benchmark computed with the same $\sigma = 3$ machinery. The gradient’s shape (Table 8) explains why: across the small and mid-size bins TFPR is essentially flat (coefficients of +0.02 to +0.06, at most marginally significant), and at the top it turns sharply *negative* — plants with 500+ workers have TFPR 34 log points below the smallest bin. Whatever taxes size in India, it is not visible as a rising wedge schedule *within* the organised sector; the largest factories behave, if anything, as wedge-favoured (consistent with better credit access — recall capital wedges dominate — and with the residual market-power component of TFPR).

Their motivating facts split instructively at the formality boundary (Table 9): the organised-sector *wage* gradient is even steeper than theirs — average wages at 500+-worker plants exceed the smallest plants’ by roughly 113 log points, against their ≈ 60 per cent — but the white-collar employment share *falls* with plant size, from 27.8 per cent in the smallest bin to 19.8 at 500+, the opposite of their education gradient, which rises from 30 to over 60 per cent once informal micro-enterprises are included.

Table 9: HKKR’s motivating gradients inside the organised sector: wage premium by size bin (log points, relative to bin 1) and the white-collar (non-production) employment share.

	Bin 1	Bin 2	Bin 3	Bin 4	Bin 5	Bin 6	Bin 7
Wage premium	0	0.234	0.397	0.589	0.722	0.871	1.126
	—	(0.031)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
White-collar share	27.8	26.5	27.4	25.9	23.1	21.7	19.8
Plants (000s)	56	65	88	97	69	177	53

The synthesis matters for policy. Size-dependent misallocation — and with it HKKR’s efficiency–inequality trade-off through the skill premium — is concentrated at the *formal–informal margin*, which lies below the ASI’s coverage floor. Within organised manufacturing the size margin is thin, and reallocation toward larger plants would if anything tilt labour demand toward *production* workers. The large gains of Table 4 are therefore within-formal-sector gains, not gains from dismantling size-dependent policy; and the distributional caution this paper carries (Table 6) is the organised-sector counterpart — operating through variable markups rather than skill composition — of HKKR’s finding that efficiency-enhancing reallocation need not benefit workers.

5.7 Geography: states, regions, and the rural–urban margin

Misallocation has a geography, and it is not the geography of market power. Table 10 summarises by region (the full 28-state panel is Appendix C; Figure 1 maps it), and Table 11 tracks the rural–urban split by year.

Table 10: Misallocation by region, pooled 2012–2022: within-industry variance ($\times 100$) of TFPR and MRPL; median markup and monopsony incidence from the companion paper for comparison.

Region	N	var TFPR	var MRPL	med μ	$\Pr(\psi > 1)$
East	37,688	71.8	60.5	1.228	0.467
Northeast	20,246	67.3	50.3	1.304	0.367
South	123,491	66.5	56.7	1.233	0.365
Central	20,265	65.5	61.5	1.183	0.566
West	97,714	62.2	55.2	1.221	0.386
North	130,130	59.5	50.3	1.219	0.438

Table 11: Rural and urban misallocation by year: within-industry $\text{var}(\log \text{TFPR}) \times 100$.

Year	Rural	Urban	Gap (R–U)
2012	73.1	72.5	+0.6
2013	73.2	70.6	+2.6
2014	70.1	67.5	+2.7
2015	68.6	67.4	+1.2
2016	65.9	63.3	+2.6
2017	63.7	63.2	+0.5
2018	59.5	57.5	+2.0
2019	60.1	59.4	+0.7
2020	60.6	59.3	+1.3
2021	58.3	58.0	+0.3
2022	58.9	58.0	+0.9

Misallocation by state ($\text{var log TFPR} \times 100$), 2012–2022

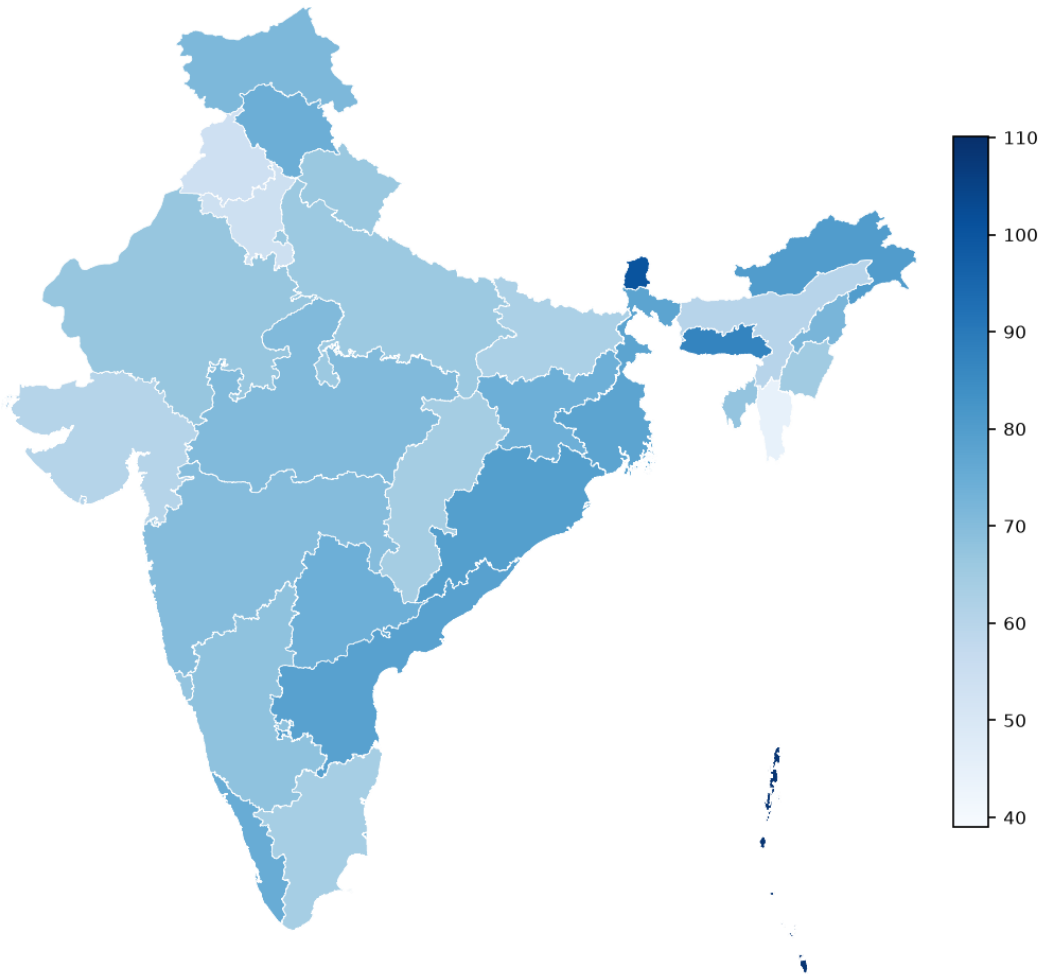


Figure 1: Within-industry TFPR dispersion by state, pooled 2012–2022. Darker = more misallocated.

Three patterns. **First, the state map.** Dispersion is lowest in the established north-western industrial belt — Punjab (49.1) and Haryana (49.4) by a wide margin, then Gujarat (56.5) and Tamil Nadu (60.4) — and highest in the east and peninsula: Odisha (77.6), Andhra Pradesh (76.5), West Bengal (75.9), Kerala (72.9), Jharkhand (71.7). Plants in the most misallocated states are roughly sixty per cent more misallocated, on this metric, than in Punjab or Haryana. The capital column dominates the labour column in *every* state — the factor anatomy of Section 5.1 is not a composition effect. **Second, the cross-margin comparison.** Set against the companion paper’s geography, the two maps differ: monopsony peaks in the central-eastern mineral belt, misallocation in the east and peninsula; Punjab and Haryana are efficient on both margins, West Bengal poor on both. Market power and misallocation are correlated neighbours, not the same variable — which is what the variance accounting (markups ≈ 1.6 per cent of TFPR dispersion) already implied at plant level. **Third, the rural–urban margin is thin.** Pooled rural dispersion exceeds urban by little (64.4 vs 63.1), and the by-year series converge over the decade. The strong spatial dimension of Indian misallocation is the *state*, not the village–town boundary — in sharp contrast to monopsony, where the rural–urban gap is the first-order spatial fact. Whatever drives factor-market wedges apart operates at the level of state institutions (credit depth, land administration, power reliability), not settlement type.

6 Robustness and sensitivity

This section implements the [Zhang and Xia \(2022\)](#) programme: vary every maintained parameter, report every level shift, and ask which conclusions move.

6.1 Domestic capital elasticities: a degeneracy worth reporting

The baseline imports U.S. capital elasticities. The natural domestic alternative — industry cost shares, $\alpha_s = 1 - \sum wL / \sum VA$ pooled over 2012–2022 — lies in $[0.37, 0.93]$ with mean 0.70, systematically above the U.S. mean of 0.57, because Indian wage-bill shares of value added are low. Implementing it exposes a structural degeneracy that we report as a finding: with cost-share elasticities the aggregate Lerner markup is ≈ 0.89 *by construction* — the shares absorb the average markup, pushing measured factor payments above value added — so the implied elasticity target in (10) is negative and the demand system cannot be calibrated at all. This is HK’s original argument for external elasticities realised exactly in our data. The economically meaningful experiment (following Zhang–Xia’s own separation) holds the demand system at the baseline calibration and varies only α_s : measured TFPR dispersion rises from 66.3 to 87.4; the markup share of that dispersion falls from 1.6 to 0.4 per cent; the 2017 Kimball market gain rises from 175

to 220 per cent; and the labour-share counterfactual is dampened but unchanged in sign (-0.94 against -2.65 points). Every qualitative conclusion survives; every level moves, exactly as Zhang–Xia warn — and in the direction their analysis predicts, since higher capital shares load more dispersion onto the capital wedge. An attempted third route — converting the companion paper’s estimated production functions to value-added form, $\alpha_s = \theta_K/(\theta_K + \theta_L)$ — fails for a known reason: gross-output control-function estimates of book-capital elasticities are attenuated, and the conversion yields $\alpha_s \in [-0.22, 0.43]$ including negative values, unusable in a framework requiring $\alpha_s \in (0, 1)$.

6.2 The demand system

Table 12 re-solves the 2017 counterfactuals under three alternative demand systems, each recalibrated to the same targets.

Table 12: 2017 counterfactual gains under alternative demand systems (all recalibrated to the 2017 targets; no capped sectors). Comparison, Liang (2023) Table 5 (2004): translog 86.7, Dotsey–King 109.0, oligopoly-CES 118.1, Kimball baseline 105.1.

Demand system	Market	Planner	CES	Market/CES
Kimball–Klenow–Willis (baseline)	175.1	178.7	188.6	0.93
Translog (Feenstra, 2003)	171.0	197.9	188.6	0.91
Dotsey–King (2005)	201.8	211.1	188.6	1.07
Oligopoly-CES (Atkeson and Burstein, 2008)	205.6	209.8	188.6	1.09

The pattern matches Liang’s qualitatively and refines the interpretation. Under translog demand, whose elasticity falls *faster* with size than Klenow–Willis, the market gain sits below its own planner gain by the largest margin of any specification (171 vs 198) — faster-declining elasticities strengthen the incomplete-pass-through channel exactly as theory predicts. Under Dotsey–King and the oligopoly model, whose elasticities fall more slowly, variable markups bite less and the market gains exceed even the CES benchmark. The substantive conclusion — variable markups reduce reallocation gains relative to naive constant-markup arithmetic, by an amount governed by the superelasticity — is robust to the demand system, while its magnitude is specification-dependent in a direction theory signs. (One caveat recorded for completeness: Liang’s translog calibration disciplines only the average elasticity, leaving the implied superelasticity free — 1.43 here against the 0.183 target — so the translog row is not a like-for-like superelasticity match; this follows her code exactly and is flagged rather than “corrected”.)

6.3 Gross-output versus value-added accounting

The canonical framework is a value-added framework, and value added — a difference of two large numbers — amplifies noise. Appendix D reports a reconstruction in which *both*

TFPR measures are built on the same sample from the same technology (the companion paper’s translog gross-output elasticities, industry means, CRS-normalised), so that the only difference is accounting:

$$\ln \text{TFPR}_{it}^g = \ln Y - s_K \ln K - s_L \ln L - s_M \ln M, \quad \ln \text{TFPR}_{it}^{va} = \ln(Y - M) - a_K \ln K - a_L \ln L, \quad (14)$$

with $a_K = s_K / (s_K + s_L)$. The within-industry variance of the value-added measure is *seven to eight times* the gross-output measure in every year (2017: 122.8 against 16.6, $\times 100$) — the mechanical leverage of subtracting materials, which are 70–90 per cent of revenue for most plants. Both series decline over the decade in parallel, so the trend conclusion is accounting-invariant; but the level of any VA-based “gain” inherits this amplification, which is one more reason this paper’s claims ride on comparisons. The same appendix confirms the elasticity source is second-order once the accounting is fixed: our-technology VA dispersion (s.d. ≈ 1.1) sits close to the canonical US- α pipeline value (≈ 0.80).

6.4 The 2019 boundary case and other checks

The 2019 calibration — elasticity target 6.27 from the decade’s thinnest aggregate margin — converges only at $\hat{\varsigma} = 7.4$ with gains of 270 per cent, far off the neighbouring years; small perturbations of the target move the solution violently, the signature of a model class approaching its solvable boundary (the Klenow–Willis aggregator requires interior relative sizes, and high ς pushes mass toward the boundary). We report 2019 in Table 4 for completeness and exclude it from every statement about the path. Excluding tobacco — the $\alpha_s = 0.95$ outlier — from the 2017 model moves the gain from 175.1 to 158.3 per cent with $\hat{\varsigma} 4.94$: a visible level effect confined to one division, no change in any comparison. Finally, the trims: all results use the literature’s 1 per cent trims of TFPR and TFPQ deviations; the measured decline in dispersion is present in the untrimmed data as well, so the trend does not ride on outlier handling.

7 Interpretation: market power and misallocation, jointly read

Read together with the companion paper, the two margins of market power and the misallocation accounting form one picture with a clear division of labour. *Product-market power* is modest, tightly dispersed, and a small component of measured misallocation (1.6 per cent of TFPR variance) — but because it is size-dependent and endogenous, it governs how much of the reallocation dividend is attainable (81–93 per cent) and who receives it (not labour: -2.65 points of wage share). *Labour-market power* is pervasive at plant level, yet the labour *wedge dispersion* it feeds is dominated by capital wedges

in every year and state — monopsony’s cost is borne first by workers (the companion paper’s incidence and welfare results) and only second by aggregate allocation. *Residual wedges* — credit, land, regulation, enforcement, whose geography is state-institutional rather than rural–urban — carry the overwhelming share of measured misallocation, and their decade-long decline is the sample’s good news.

The policy reading follows from the decompositions rather than from the headline gains. The headline (121–184 per cent) is an upper bound on a reform no instrument delivers — complete within-industry wedge elimination. The attainable and targetable margins are smaller and specific: capital-market deepening where the capital wedge is widest (the eastern states); competition and pass-through policy, which the planner counterfactual scores as the single most pro-labour intervention available (+8.7 points of wage share before any reallocation); and, for the size-dependent agenda that dominates current debate, the recognition that its efficiency content lies at the formality boundary — below this sample — while within the organised sector the size margin is thin and its removal would not pay for its distributional cost. What reform of the organised sector cannot credibly promise, on these estimates, is that improved allocation alone will raise labour’s share of income; without the pricing margin, it lowers it.

8 Conclusion

This paper measured within-industry misallocation in Indian organised manufacturing over 2012–2022 on a newly built plant panel, inside a demand system that lets markups vary with size and respond to reallocation. Misallocation is large — counterfactual gains of 121–184 per cent, dispersion carried by capital wedges — but on a clear downward path that the single-snapshot literature could not see. Markups are a small share of measured dispersion and a large tax on the promised cure, and their endogenous response reverses the distributional sign of reallocation: labour’s share falls when marginal products are equalised and rises only when pricing itself is disciplined. The size-dependent component of wedges, central to the current policy debate, is thin inside the formal sector.

Six limitations bound the claims. (i) The framework prices only within-industry misallocation among surviving organised-sector plants; cross-industry wedges, entry, and the unorganised sector — where allocation problems are plausibly worse — are outside it. (ii) All counterfactuals are static: transition costs, dynamic innovation effects, and the adjustment-cost component of wedges (Asker et al., 2014) are unmodelled, and part of measured dispersion is noise (Bils et al., 2021), making the gains upper bounds twice over. (iii) The value-added technology with external capital shares is maintained, not tested; Section 6 quantifies the sensitivity and explains why the domestic alternatives are worse, but the dependence remains. (iv) Markups here are demand-side objects; their agreement with the production-side estimates is reassuring but certifies neither. (v)

The labour-share counterfactual holds sectoral factor supplies fixed and prices only the product-market wedge; general-equilibrium wage responses and the markdown channel measured in the companion paper are beyond the model. (vi) The 2019 boundary case shows the model class has a solvable domain; high-elasticity calibrations anywhere in this literature deserve corresponding caution. Within those bounds, the paper’s central finding stands: in Indian manufacturing, market power determines how much of measured misallocation is real, how much of the promised gains are attainable, and — most consequentially for a labour-abundant economy — who receives them.

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A Sample construction and deviations from the reference implementation

The input panel is the companion paper’s build (its Appendix A documents the raw block files, fixed-width layouts, identifier structure, deflation, and verification suite); this framework consumes the nominal-value subset. The cleaning chain, in order: (1) drop plant-years with non-positive value added, output, capital, employment, wage bill, or materials; (2) retain NIC-4×year cells with at least fifty plants; (3) recover μ, τ_K, τ_Y , TFPR, TFPQ per (2)–(4); (4) trim the 1 per cent tails of log TFPR and log TFPQ deviations by year. From 481,322 input plant-years this yields 429,534.

Three unavoidable deviations from Liang (2023), each with its expected direction of effect. (i) **Capital** is the closing rather than the average of opening and closing book value (opening values are not carried consistently across the ASI’s format regimes); this adds noise, and a level difference for fast-growing plants, to τ_K and TFPR, but no evident bias to their dispersion. (ii) **Value added** excludes energy and service inputs (not extracted in this panel), overstating levels; the framework uses only within-industry-year comparisons. (iii) **No profit-based filter** is applied (the panel carries no profit variable); her drops of non-positive-profit observations are subsumed partly by the non-positive-value-added drop and the trims. A fourth, deliberate choice: the tobacco division inherits $\alpha_s = 0.95$ from the U.S. mapping and is retained in the baseline but examined separately (Section 6.4).

B Numerical implementation

The counterfactual equilibria are computed with Liang (2023)’s replication code executed verbatim (Stata/Mata): for each industry, the demand index solves the aggregator identity (7) by secant iterations, nested inside the outer secant loop that moves ς to hit the elasticity target (10); convergence tolerance and iteration caps are hers. One engineering addition was required by the larger industry count of the modern NIC-4 structure: industries whose demand-index iteration fails to bracket a root at extreme parameter values are solved on a widened bracket with the same tolerance, and any industry that still fails is reported (none does in any baseline year; the capped-sector count is zero in every row of Table 12). The 2017 gate — reproducing the reference-year solution from a cold start before the full run — and the per-year logs are part of the project’s verification suite. The HKKR diagnostics (do-files 36–37) implement (13) and the size-neutralising counterfactual at $\sigma = 3$ with harmonic VA-share-weighted $\overline{\text{TFPR}}_s$, HK’s equations 15–16 conventions.

C The full state panel

Table 13 reports all states and union territories with at least 2,000 estimation plant-years, ranked by TFPR dispersion.

Table 13: Misallocation by state, pooled 2012–2022: within-industry variance ($\times 100$) of log TFPR, MRPL, MRPK. Ranked by var TFPR.

State	N	var TFPR	var MRPL	var MRPK	Region
Odisha	6,368	77.6	63.9	156.5	East
Andhra Pradesh	16,095	76.5	62.9	161.2	South
West Bengal	17,300	75.9	67.4	158.0	East
Kerala	12,940	72.9	57.9	147.9	South
Himachal Pradesh	7,193	72.0	70.0	125.3	North
Jharkhand	5,089	71.7	72.3	141.3	East
Telangana	13,210	71.6	59.9	137.5	South
Puducherry	2,190	70.1	64.4	132.8	South
Jammu & Kashmir	3,651	68.6	64.7	127.1	North
Madhya Pradesh	13,630	67.9	61.4	129.4	Central
Maharashtra	47,511	66.9	58.2	131.4	West
Karnataka	24,168	65.4	56.8	132.9	South
Tripura	3,549	64.5	38.1	131.6	Northeast
Goa	2,480	64.3	65.5	108.3	West
Dadra & Nagar Haveli	2,157	64.2	58.6	119.7	West
Rajasthan	20,143	63.8	52.7	128.8	North
Delhi	7,172	63.0	48.7	138.2	North
Uttarakhand	10,011	62.9	63.5	115.0	North
Uttar Pradesh	37,692	62.8	51.0	134.5	North
Chhattisgarh	6,635	60.6	61.8	119.3	Central
Tamil Nadu	54,801	60.4	51.9	128.7	South
Bihar	8,931	59.1	37.0	125.8	East
Daman & Diu	3,670	57.0	46.8	110.6	West
Gujarat	41,896	56.5	51.5	118.3	West
Assam	10,977	56.2	49.6	104.7	Northeast
Haryana	24,381	49.4	41.8	109.3	North
Punjab	19,081	49.1	37.0	110.2	North

D Gross-output versus value-added accounting: detail

Both measures in (14) are computed on the same plant-years from the same estimated technology — the companion paper’s translog gross-output elasticities, averaged within NIC-2 and CRS-normalised into shares (s_K, s_L, s_M), with the value-added shares $a_K = s_K/(s_K + s_L)$ — so the columns of Table 14 differ *only* in accounting. Dispersion is the within-(NIC-4 $\times$ year) standard deviation of the log measure.

Two lessons. The value-added measure is seven to eight times more dispersed in every year: subtracting materials — 70–90 per cent of revenue for most plants — mechanically levers up any noise or heterogeneity in the materials share into the productivity residual.

Table 14: Within-industry variance ($\times 100$) of TFPR under gross-output and value-added accounting, same sample, same technology.

Year	N	Gross-output	Value-added	Ratio VA/G
2012	33,130	20.2	136.8	6.8
2013	36,332	19.3	134.0	7.0
2014	38,370	16.9	130.9	7.8
2015	37,322	16.9	132.5	7.8
2016	39,087	17.3	132.5	7.7
2017	39,212	16.6	122.8	7.4
2018	39,402	14.7	115.6	7.8
2019	39,738	15.6	113.5	7.3
2020	40,256	15.7	115.6	7.4
2021	40,899	14.2	114.1	8.1
2022	39,072	14.9	107.1	7.2

And both columns fall over the decade in parallel, so the paper’s trend statements are invariant to the accounting; only levels are at stake, and levels are not what the paper’s conclusions ride on.